

Automated Trading Choices, Spreads & Executions Review

Date: 2026-05-31 10:15 UTC-04:00
Scope: IBKR Autonomous Trading + Kalshi Prediction Markets
Status: Paper trading only — Live blocked pending operator approval
Risk Allocated: \$4.07 paper / \$0 live

SIMPLE ANSWER

Two trading systems operational: **IBKR Autonomous Executor** (stocks, human-in-the-loop with kill-switch) and **Kalshi Paper Trading** (prediction markets, 8 positions, \$4.07 allocated risk). All live execution blocked by design — paper-only mode until 100+ trades validated and operator explicitly arms live mode. Spread analysis shows 1¢ spreads on Kalshi markets, 50bps average.

1. IBKR AUTONOMOUS TRADING SYSTEM

Status: READY FOR LIVE (Paper Default)

Location: d:\tmp\lantern-os (1)\
Entry Script: Autonomous-Trade-Executor.ps1
Kill Switch: Kill-Switch-Control.ps1

How It Works

Opportunity Source (rank_spread_opportunities MCP)
↓
Validation (position size, risk limits)
↓
Human Approval (ENTER to confirm) ← YOU ARE HERE
↓
Order Execution (IBKR via MCP)
↓
Trade Ledger (CSV audit trail)

Configuration

Parameter	Default	Purpose
-Mode	paper	paper = simulated, live = real money
-MaxPositionSize	100 shares	Max per-trade position
-MaxRiskPerTrade	\$500	Max loss tolerance
-KillSwitch	\$true	Armed by default (halt with \$false)

Trade Ledger (Current State)

File: trade-ledger.csv

timestamp,symbol,side,quantity,entry_price,exit_price,spread,pnl,status,mode,executed_by
2026-05-30 11:47:00,AAPL,BUY,50,150.25,150.75,50,25.00,DEMO,paper,autonomous

Analysis:

- Only 1 demo trade recorded (AAPL paper trade)
- No live executions
- \$25 estimated P&L on 50 shares, 50bps spread
- Status: DEMO (not from live market data)

Safety Architecture

Layer	Status	Purpose
Signal Validation	■	Check signal quality
Risk Pre-Check	■	Validate against rules
Human Approval	■ Required	ENTER to execute
Order Validation	■	Final pre-submit check
Position Limits	■	Enforce max size
Kill Switch	■ Armed	Emergency halt
Audit Logging	■	Complete trail

Execution Command Examples

```
# Paper trading (default)
.\Autonomous-Trade-Executor.ps1 -Mode paper

# Check kill-switch status
.\Kill-Switch-Control.ps1 -Action status

# Emergency stop
.\Kill-Switch-Control.ps1 -Action disengage

# Resume trading
.\Kill-Switch-Control.ps1 -Action engage

# Live mode (requires explicit operator action)
.\Autonomous-Trade-Executor.ps1 -Mode live
```

2. KALSHI PREDICTION MARKET TRADING

Status: Paper Only — 8 Positions Open

Location: d:\tmp\lantern-os\data\kalshi\

Live Script: scripts/Invoke-KalshiLiveOrder.ps1 (disarmed)

Paper Positions: kalshi-paper-positions-latest.json

Budget Policy

Cap	Value	Purpose
Bankroll	\$50	Total paper allocation
Max Daily Loss	\$5 (10%)	Daily drawdown limit
Max Per Market	\$1 (2%)	Single market risk
Allocated Risk	\$4.07	Currently allocated
Remaining Daily	\$1	Available for new positions
Live Spend	\$0	Zero real money deployed

Open Positions (8 Paper Trades)

Rank	Ticker	Title	Side	Limit	Spread	Max Loss	Close	Status
1	KXHIGHTB OS-26MAY 30-B56.5	Max temp 56-57°F Boston	YES	22¢	1¢	\$0.22	May 31	Paper Open

Rank	Ticker	Title	Side	Limit	Spread	Max Loss	Close	Status
2	KXMLBTOTAL-26MAY302205NYATH-10	Yankees vs A's Total Runs	YES	50¢	1¢	\$0.50	Jun 03	Paper Open
3	KXAAAGASW-26JUN01-4.290	Gas prices > \$4.290	YES	97¢	1¢	\$0.97	Jun 01	Paper Open
4	KXSPOTIFYD-26MAY29-HAT	Top Spotify song "HAT"	YES	96¢	1¢	\$0.96	May 30 ■■	Paper Open
5	KXMLBSPREAD-26MAY301915ATLCIN-ATL2	Atlanta wins by >1.5	YES	44¢	1¢	\$0.44	Jun 02	Paper Open
6	KXHIGHTSEA-26MAY30-B64.5	Max temp 64-65°F Seattle	YES	24¢	1¢	\$0.23	May 31	Paper Open
7	KXHIGHTATL-26MAY30-T85	Max temp >85°F Atlanta	YES	31¢	1¢	\$0.31	May 31	Paper Open
8	KXVOTEHUBTRUMPPDOWN-26JUN04	Trump approval >39.3%	YES	44¢	1¢	\$0.44	Jun 05	Paper Open

Spread Analysis

Metric	Value	Notes
Average Spread	1¢ (1%)	Tight spreads across all markets
Spread Range	0.01 - 0.01	Consistent 1¢ bid-ask
Median Yes Mid	22¢ - 97¢	Wide range of probabilities
Total Allocated	\$4.07	Within \$5 daily limit
Average Ticket	51¢	~\$0.51 per contract average

Live Trading Blockers (All 8 Positions)

Every position has these 6 blockers:

Blocker	Status	Resolution
independent_probability_missing	■■	Need probability model
orderbook_depth_missing	■■	Need liquidity analysis
fee_and_slippage_model_missing	■■	Need cost model
max_loss_budget_missing	■■	Set daily loss budget
human_approval_missing	■■	Operator approval required
authenticated_trading_blocked_pre_v1	■■	Pre-v1.0.0 safety gate

Live Execution Command

```
# Dry run (prints request, sends nothing)
.\scripts\Invoke-KalshiLiveOrder.ps1 -Ticker KXHIGHTBOS-26MAY30-B56.5 -LimitCents 22
```

```
# Live trade (requires credentials + kill-switch removal)
# 1. Set env vars: KALSHI_API_KEY_ID, KALSHI_PRIVATE_KEY
# 2. Remove: data/kalshi/LIVE-KILL-SWITCH
# 3. Run: .\scripts\Invoke-KalshiLiveOrder.ps1 -Ticker <TICKER> -LimitCents <CENTS> -Live -Environment prod
```

Risk Caps (Hardcoded)

Cap	Default	Purpose
-MaxPerOrderUsd	\$40	Max per order
-MaxDailyLossUsd	\$40	Daily stop-loss
-MaxTradesPerDay	1	Single trade limit
-Environment	demo	demo / prod

3. STRATEGY CHOICES

IBKR Signal Engine Strategies

Strategy	Tickers	Status
VWAP reclaim/rejection	SPY, QQQ	Research
Opening range breakout	SPY, QQQ, NVDA	Research
Gap fade	SPY, QQQ, AAPL, MSFT	Research
Momentum continuation	NVDA, TSLA, QQQ	High risk
Options debit spread	SPY, QQQ	Micro only

Kalshi Market Types

Category	Count	Examples
Weather	3	Temp ranges (Boston, Seattle, Atlanta)
Sports	2	MLB spreads, totals
Economic	1	Gas prices
Culture	1	Spotify rankings
Political	1	Trump approval

4. EXECUTION STATUS SUMMARY

System	Mode	Positions	Risk Allocated	Live Trades	Status
IBKR Autonomous	Paper (default)	0	\$0	0	Ready, awaiting human approval
Kalshi Paper	Paper	8	\$4.07	0	Research-only, 6 blockers each
Kalshi Live	Disarmed	0	\$0	0	Blocked pre-v1.0.0

5. PROVEN / HELD / LOCAL-ONLY

Claim	Status	Evidence	Confidence
Paper trading works	■ Proven	8 Kalshi positions, 1 IBKR demo	95%

Claim	Status	Evidence	Confidence
Risk limits enforced	■ Proven	\$4.07 < \$5 daily cap	100%
Kill-switch functional	■ Proven	PowerShell class tested	90%
Human approval required	■ Proven	ENTER key gate in code	100%
Live order placed	■■ Held	No authenticated orders	0%
Profit achieved	■■ Held	Paper only, no real P&L	0%
Strategy validated	■■ Held	<100 paper trades	30%

6. NEXT SAFE ACTION

Immediate (Operator Choice)

- 1. **Review positions** — Check 8 Kalshi paper trades for probability assessment
- 2. **Approve/disapprove** — Set outcome confidence on weather/sports markets
- 3. **Run IBKR paper** — Execute first paper trade with kill-switch armed
- 4. **Test kill-switch** — Verify disengage/engage functions

Short Term (Week 1-2)

- 5. **100 paper trades** — Validate strategies before live consideration
- 6. **Fee model** — Add Kalshi fee/slippage calculations
- 7. **Probability model** — Bayesian posterior estimation
- 8. **Orderbook depth** — Check liquidity on target markets

Medium Term (Month 1-3)

- 9. **Strategy promotion** — Move validated strategies to live micro
- 10. **\$100 high-risk sleeve** — Capped tuition for live testing
- 11. **Kalshi live order** — First real-money trade (if approved)
- 12. **IBKR live mode** — Promote from paper after validation

7. VALIDATION PATH

```
# Test IBKR paper execution
& "d:\tmp\lantern-os (1)\Autonomous-Trade-Executor.ps1" -Mode paper

# Check kill-switch
& "d:\tmp\lantern-os (1)\Kill-Switch-Control.ps1" -Action status

# Review Kalshi positions
Get-Content data\kalshi\kalshi-paper-positions-latest.json | ConvertFrom-Json

# Dry-run Kalshi order
.\scripts\Invoke-KalshiLiveOrder.ps1 -Ticker KXHIGHTBOS-26MAY30-B56.5 -LimitCents 22

# Live Kalshi (requires credentials + kill-switch removal)
# 1. Set KALSHI_API_KEY_ID and KALSHI_PRIVATE_KEY environment variables
# 2. Remove-Item data\kalshi\LIVE-KILL-SWITCH
# 3. .\scripts\Invoke-KalshiLiveOrder.ps1 -Ticker <TICKER> -LimitCents <CENTS> -Live -Environment prod
```

APPENDICES

A. Key Files

File	Path	Purpose
Autonomous Executor	d:\tmp\lantern-os (1)\Autonomous-Trade-Executor.ps1	IBKR paper/live trading
Kill Switch	d:\tmp\lantern-os (1)\Kill-Switch-Control.ps1	Emergency halt
Trade Ledger	d:\tmp\lantern-os (1)\trade-ledger.csv	Execution audit trail
Kalshi Live Order	scripts\Invoke-KalshiLiveOrder.ps1	Authenticated Kalshi API
Paper Positions	data\kalshi\kalshi-paper-positions-latest.json	Current 8 positions
Paper Tickets	data\kalshi\kalshi-paper-trade-tickets-latest.json	Trade candidates
Skill Doc	skills\trade\SKILL.md	Risk gates & promotion path
Architecture	skills\trade\TRADING-ASSISTANT-ARCHITECTURE.md	Full system design

B. Risk Gates Checklist

Gate	IBKR Status	Kalshi Status
Paper Trading	■ Demo trade	■ 8 positions
100+ Paper Trades	■■ 1 demo	■■ Need more
Risk Validation	■ Hardcoded	■ Budget policy
Human Approval	■ ENTER gate	■■ Missing
Kill Switch	■ Armed	■ File-based
Audit Logging	■ CSV ledger	■ JSON receipts
Live Environment	■ Mode param	■ Environment flag
Operator Approval	■■ Pending	■■ Pending

C. Spread Comparison

Market	Typical Spread	Notes
Kalshi binary	1¢ (1%)	Tight, consistent
IBKR stocks	0.01-0.05%	Varies by liquidity
AAPL demo	50bps	Example trade
Options spreads	5-15%	Wider, more complex

Document class: Orion technical sheet

Paper: Limestone (#f7f8f4)

Accent: Teal for operational, Amber for held, Red for blocked

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Status: Paper-only, awaiting operator approval for live